1.3.2 Set Builder Form or Rule Form

In set builder form, all the elements are described by a rule.

For example,

(i) $A = \{x : x \text{ is a vowel in English alphabets}\}$

(ii) $B = \{x | x \text{ is a whole number}\}$

Note

The symbol ':' or '|' stands for "such that".

1.3.3 Roster Form or Tabular Form

A set can be described by listing all the elements of the set.

For example,

(i) $A = \{a, e, i, o, u\}$

(ii) $B = \{0,1,2,3,\ldots\}$

Can this form of representation be possible always?

Note

Three dots (…) in the example (ii) is called ellipsis. It indicates that the pattern of the listed elements continues in the same manner.

Activity-2

Write the following sets in respective forms.

S.No.	Descriptive Form	Set Builder Form	Roster Form
1	The set of all natural numbers less than 10		
2		$\{x : x \text{ is a multiple of 3}, x \in \mathbb{N}\}$	
3			{2,4,6,8,10}
4	The set of all days in a week.		
5			{…-3,-2,-1,0,1,2,3…}

Example 1.2

Write the set of letters of the following words in Roster form

(i) ASSESSMENT (ii) PRINCIPAL

Solution

(i) ASSESSMENT

$X = \{A, S, E, M, N, T\}$

(ii) PRINCIPAL

$Y = \{P, R, I, N, C, A, L\}$

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